

DS 4440 Supervised Learning (Review)

Given Training data (X, y)

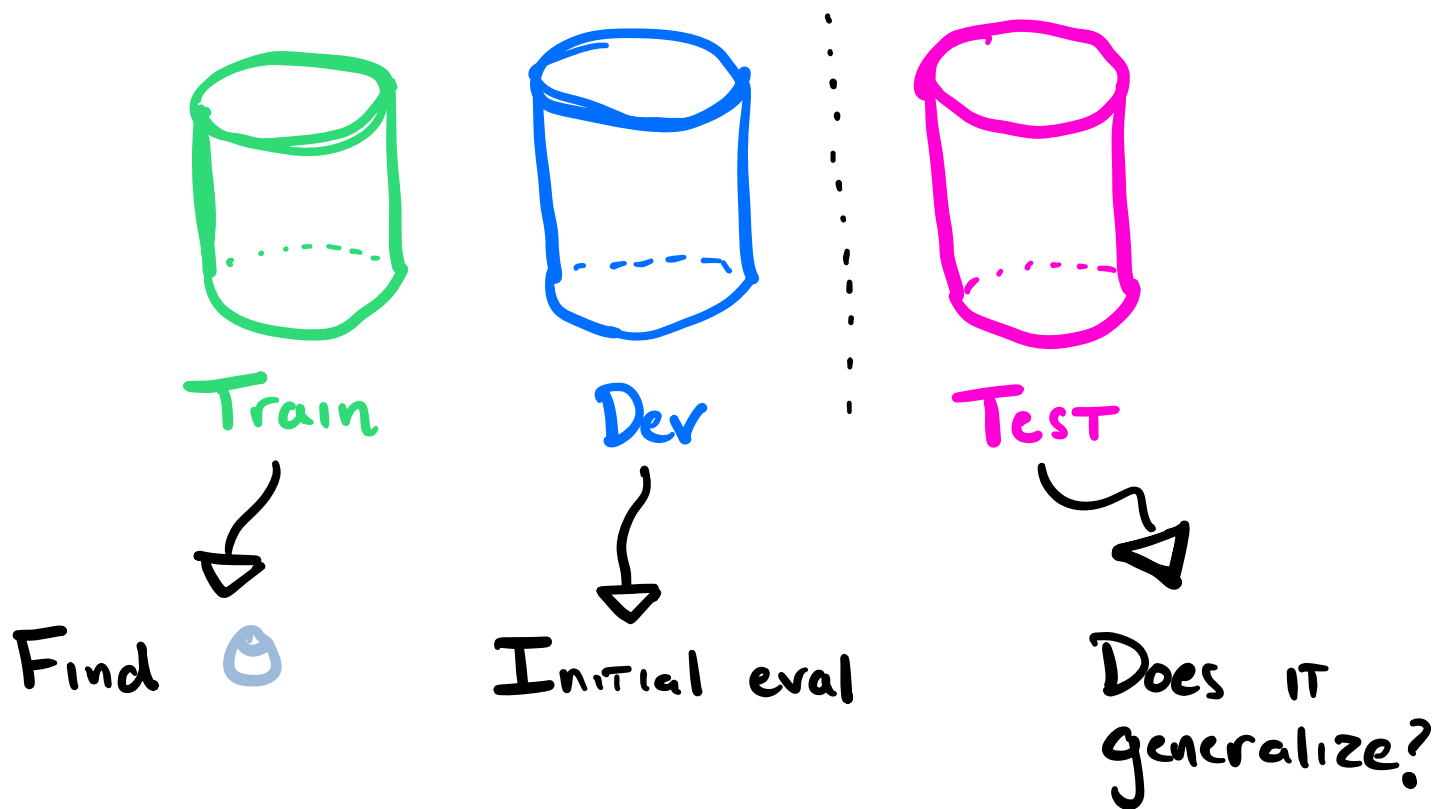
$$X = \{x_1 \dots x_N\}$$

$$= \begin{bmatrix} x_{11} & \dots & x_{1D} \\ \dots & \ddots & \dots \\ x_{N1} & \dots & x_{ND} \end{bmatrix}_N \quad X \in \mathbb{R}^{N \times D}$$

$$y = [y_1 \dots y_N] \quad y \in \mathbb{R}^N$$

The goal: find parameters θ for
function f_θ s.t. $f_\theta(X) = y$

This should generalize beyond X .



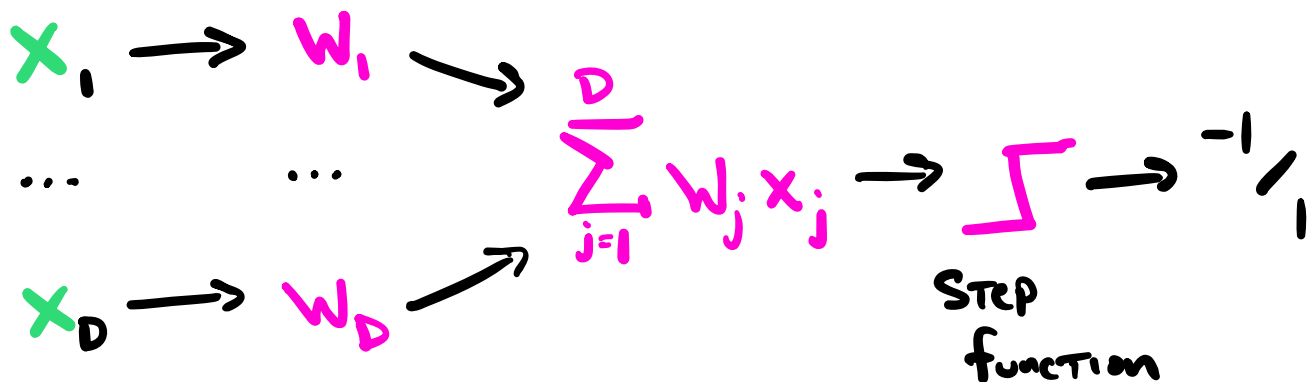
Learning means finding a  s.t.
loss on (X, y) is minimized.
($\mathcal{L}$)

$$\arg \min_{\hat{\theta}} \sum_{i=1}^N \mathcal{L}(f_{\hat{\theta}}(x_i), y_i)$$

  $\hat{y}_i$

Machine Learning

One choice The Perceptron



$$\hat{y}_i = \begin{cases} 1 & \text{if } w x_i > \tau \\ -1 & \text{otherwise} \end{cases}$$

The loss is 0/1.

$$L(\hat{y}_i, y_i) = \begin{cases} 0 & \text{if } \hat{y}_i = y_i \\ 1 & \text{otherwise} \end{cases}$$

Learning

$$W \leftarrow \emptyset$$

$$(\Theta = W)$$

for $(x_i, y_i) \in (X, Y)$

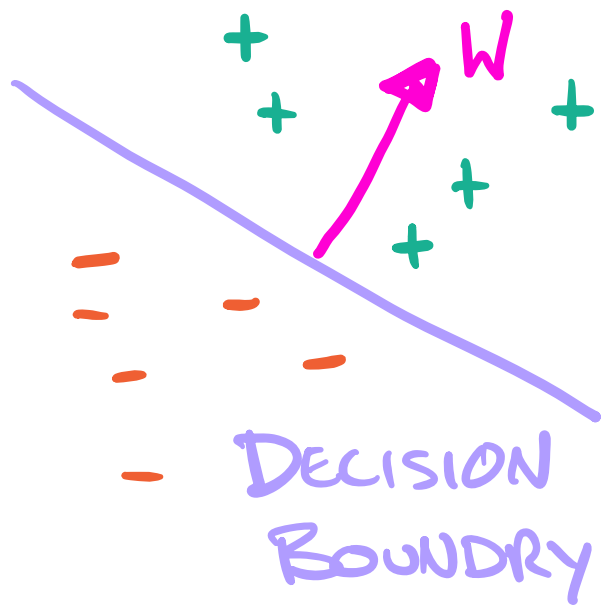
if $\text{Step}(W x_i) \neq y_i$

$$W \leftarrow W + y_i x_i$$

★ Assumes bias baked into x_i , w

(See CoLab!)

Why does This Work?!



Weight Vector
is $\perp$ To The
decision boundary.

Consider where
 $Wx_i = \phi$.

Suppose we make
an **incorrect** prediction for
 x_i That should be **+**.

$$\hat{W}' \leftarrow \hat{W} + x_i$$

So new score is

$$\begin{aligned}\hat{W}' \cdot x_i &= (\hat{W} + x_i) \cdot x_i \\ &= \hat{W} \cdot x_i + \underbrace{\|x_i\|^2}_{\text{Positive}}\end{aligned}$$